

WEST MEMPHIS, AR, USA

WMO: 722054

Lat: 35.135N

Lon: 90.234W

Elev: 214

StdP: 14.58

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	22.3	7.3	8.1	23.1	11.3	9.9	25.4	25.8	55.0	23.9	53.4	7.7	10	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	95.5	77.3	93.5	76.9	91.8	76.5	81.0	90.4	79.9	89.0	78.7	87.5	6.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.5	148.9	85.7	77.4	143.5	84.7	76.3	138.6	83.7	44.7	90.7	43.5	88.8	42.4	87.6	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.6	18.9	16.8	DB	12.7	99.0	5.2	2.4	8.9	100.7	5.9	102.1	3.0	103.4	-0.8	105.1	
			WB	11.3	82.6	4.8	1.7	7.8	83.8	5.0	84.8	2.3	85.7	-1.2	86.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	40.4	43.9	53.0	62.1	71.2	79.1	80.8	80.2	74.9	63.2	51.5	43.8
	DBStd	16.70	10.83	10.90	10.17	8.19	7.07	4.19	3.88	4.60	6.14	8.79	9.13	9.73
	HDD50	1008	333	233	86	11	0	0	0	0	0	7	92	246
	HDD65	3138	762	593	386	147	32	0	0	0	5	141	413	659
	CDD50	5422	36	61	179	373	656	874	955	935	746	418	136	53
	CDD65	2079	0	1	15	60	223	424	490	471	301	86	7	1
	CDH74	20651	0	10	84	447	1839	4441	5278	4932	2792	787	40	1
	CDH80	8805	0	2	5	78	615	1995	2383	2290	1186	247	4	0
Wind	WSAvg	7.0	8.3	8.6	8.8	8.5	7.2	6.1	5.0	4.9	5.4	6.3	7.2	7.7
Precipitation	PrecAvg	53.6	4.5	4.3	5.3	5.4	5.3	3.7	4.1	3.1	3.0	4.1	4.7	5.5
	PrecMax	69.8	7.7	10.0	12.1	16.3	11.2	7.8	9.9	5.6	10.0	10.4	11.1	10.9
	PrecMin	34.8	1.4	1.9	1.4	1.6	1.4	1.1	0.3	0.9	0.0	0.2	1.3	1.3
	PrecStd	9.7	1.5	2.2	2.5	3.2	2.5	1.9	2.6	1.3	2.4	2.8	2.8	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.3	74.6	80.4	85.6	91.2	96.3	96.9	99.1	96.7	90.9	78.4	72.4
		MCWB	63.5	64.2	65.5	67.1	73.1	76.6	77.6	76.6	75.6	72.5	65.4	66.6
	2%	DB	65.0	70.1	76.3	82.1	88.7	93.8	94.4	95.8	93.2	86.3	73.8	67.2
		MCWB	60.0	63.5	64.5	67.4	72.2	75.6	78.1	77.9	75.2	71.0	63.1	62.4
	5%	DB	61.8	66.1	72.6	79.2	86.3	92.0	92.7	93.2	90.3	82.3	70.5	63.4
		MCWB	57.6	59.3	62.3	65.9	71.3	75.3	78.2	77.5	74.3	68.1	62.1	59.0
	10%	DB	57.7	61.8	69.0	76.0	83.7	89.9	90.7	90.8	87.1	78.4	66.3	59.6
		MCWB	53.0	56.1	59.7	64.7	70.4	74.6	77.7	77.1	73.0	66.2	59.2	55.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.4	68.0	68.5	72.2	77.0	80.5	82.4	82.9	79.1	76.3	70.3	67.9
		MCDB	68.0	70.9	76.5	79.4	85.5	90.8	91.3	93.3	89.8	85.5	74.1	71.5
	2%	WB	61.8	64.7	66.1	70.0	75.4	79.1	81.2	81.1	77.7	74.2	66.7	64.1
		MCDB	64.5	68.6	73.9	77.8	84.3	89.2	90.5	90.3	88.5	82.1	71.4	66.8
	5%	WB	58.8	60.8	64.0	68.3	73.9	78.0	80.2	79.8	76.5	71.9	63.8	60.2
		MCDB	61.5	65.2	71.0	75.7	82.1	87.6	89.3	88.5	86.1	78.4	68.7	62.9
	10%	WB	53.9	57.0	61.8	66.5	72.6	76.9	79.1	78.4	75.2	69.2	60.2	56.0
		MCDB	56.8	61.4	67.4	73.8	80.3	86.0	87.7	86.8	83.2	75.2	64.9	59.0
Mean Daily Temperature Range	5% DB	MDBR	16.3	17.0	18.6	19.6	18.4	18.9	17.5	18.8	20.9	21.4	19.7	15.9
		MCDBR	19.4	21.5	21.5	22.0	21.1	21.2	19.9	21.0	23.4	24.5	23.4	19.3
		MCWBR	15.8	15.2	13.1	10.9	8.4	7.1	7.1	7.2	7.8	11.0	14.6	15.1
	5% WB	MCDBR	17.2	19.3	19.2	18.4	17.7	17.9	17.5	17.9	20.0	20.0	18.6	17.1
		MCWBR	15.5	15.5	12.6	10.7	8.5	7.7	7.2	7.8	7.8	11.2	14.7	15.0
Clear-Sky Solar Irradiance	taub	0.320	0.328	0.366	0.405	0.445	0.466	0.494	0.488	0.422	0.361	0.334	0.318	
	taud	2.487	2.461	2.382	2.267	2.214	2.169	2.126	2.123	2.302	2.454	2.481	2.510	
	Ebn at Noon	280	291	287	279	268	262	253	252	265	275	272	274	
	Edh at Noon	27	30	36	43	45	48	49	49	39	30	27	25	
All-Sky Solar Radiation	RadAvg	772	938	1304	1662	1868	2077	1988	1833	1559	1215	862	664	
	RadStd	65	127	116	110	114	123	98	119	114	146	90	68	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	+2.39	+2.92	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page